Traceability matrix (Golden Thread)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Users need to manage their personal information, and account securely, and basic services take time to process managing items.	**FR-1** Create, edit, delete account, and upgrade to Premium	• User account entity • Registration form • Authentication • Account management • Premium Subscription	• Registration page • Login page • Account Settings	TC-01: Register, log in, edit account details and verify Premium access.
Users forget what groceries they have and may waste food or consume expired food.	**FR-2** Add, edit, and delete grocery items and mark them as used.	• Grocery item entity • Expiration date field • CRUD operations • Item status (active/used) (Ch 5 & 7)	• Pantry Dashboard • Add/Edit Grocery page	TC-02: Add, edit, delete and mark grocery items as used.
Users may find it difficult to remember or estimate expiration dates for different types of groceries.	**FR-3** Automatically calculate and display the estimated expiration date based on the grocery category and purchased date.	• Category-based shelf-life period • Calculate expiration date using purchase date • Allow date editing	• Add/Edit Grocery page	TC-03: Select a grocery category and purchase date; verify the estimated expiration date.
Users may forget about groceries until they expire, which can lead to food waste and possible health risks from consuming expired food.	**FR-4** Send expiration alerts/ notifications.	• Expiration-date checking • Reminder settings • Notification service	• Reminders page • Notification interface	TC-04: Check whether expiration reminders follow the configured reminder period.
Users may not remember what groceries they currently have at home.	**FR-5** Display all active grocery items with details.	• Query active items • Display grocery name and expiration date • Display item status	• Pantry Dashboard • Active Items section	TC-05: Add grocery items and verify that all active items appear with correct details.
Users may overlook groceries that are about to expire or have already expired.	**FR-6** Show items that are expiring soon and already expired.	• Filter items by expiration date • Identify expiring-soon items • Identify expired items	• Pantry Dashboard • Expiring Soon section	TC-06: Test different expiration dates and verify the correct item status.

		• Sort by expiration date	• Expired Items section	
Manually adding many grocery items after shopping can take too much time.	**FR- 7** Premium receipt scanning	• Camera scan •Premium-only access check •Review and confirm screen	• Premium Features page • Scan Receipt page	TC-07: Scan a sample receipt and verify that detected items are displayed for confirmation
Categorizing many grocery items individually can take additional time after shopping	**FR-8** Automatic category detection	• Automatic category detection • Suggested category per item • Confirm/edit category	• Scan Review page • Add/Edit Grocery page	TC-08: Check whether detected grocery items receive appropriate category suggestions.
Different users may need different amounts of time to use their groceries before they expire	**FR-9** Premium custom reminder settings	•Reminder-period setting •Saved user preference •Notification service	• Premium Features page • Reminder Settings page	TC-09: Change the reminder period and verify that the selected setting is saved.
Users may want to review previously expired groceries to understand their food-waste patterns.	**FR-10** Expired grocery history(Free: past 7 days, Premium:Unlimited)	• Expired-item history query • 7-day limit for Free users • Unlimited history for Premium users	• Grocery History page • Premium History section	TC-10: Check expired-item history access for Free and Premium accounts.
Users may want to understand how much food they use on time and how much they waste	**FR-11** Premium usage summary (daily, weekly and monthly).	• Usage statistics calculation • Daily, weekly and monthly summaries • Bar chart visualization • Monthly count reset	• Usage Summary page • Premium Features page	TC-11: Verify daily, weekly and monthly usage counts using sample grocery records.